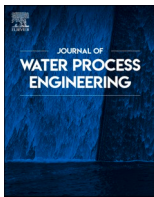




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Interpretable prediction of coagulant dosage in drinking water treatment plant based on automated machine learning and SHAP method

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ABSTRACT

Accurate prediction of coagulant dosing is critical to ensure the efficiency and cost-effectiveness of the water treatment process. This study developed an accurate prediction model for coagulant dosing in drinking water treatment plants using automated machine learning (AutoML) and enhanced its transparency with the SHAP (Shapley Additive explanation) method. Results showed that the Random Forest (RF) model developed by AutoML outperformed the best single Gradient Boosting Tree model, with a 37 % lower RMSE of 0.89, a 52 % lower MAE of 0.47, and a 5 % higher R^2 of 0.96. The RF model has the potential to save 10.25 % of coagulant dosage per year, specifically, 222 kg/d of PACl at a cost of 180.7 yuan/d for Yangtze River water, and 225 kg/d at a cost of 183.2 yuan/d for Luan River water, while ensuring a more stable water quality in the treated water. SHAP analysis identified conductivity, ammonia nitrogen, chemical oxygen demand, and temperature of raw water as key factors affecting PACl dosage, which significantly impacts treated water pH. The combination of AutoML and SHAP method applied to intelligent water management, can bolster water treatment efficiency and management practices, resulting in noteworthy environmental, economic, and social benefits.

1. Introduction

With the rapid development of water treatment technologies, the issues of energy consumption and carbon emissions are becoming increasingly prominent [1,2]. In order to achieve the goal of green and low-carbon, traditional processes have been continuously optimized, but they are still subject to technical and economic constraints: adsorption technology is difficult to regenerate; photocatalytic deep mineralization is limited by the efficiency; and microwave catalytic efficiency is high, but the cost of the equipment is high [3–6]. Meanwhile, the complexity of the water treatment process places higher demands on the chemical dosing accuracy, especially in coagulation, whose operating cost accounts for a high proportion of the total cost of water plants [7,8], and multivariate (water temperature, pH, etc.), nonlinear dynamic response, and real-time disturbances make it difficult for traditional empirical models to balance dosing accuracy and energy efficiency [9,10]. Therefore, determining the optimal coagulant dosage

based on raw water quality and established treated water quality criteria is critical to ensure water treatment effectiveness and to realize economic and environmental benefits [11].

Traditional manual dosing is highly subjective, resulting in high drug consumption and difficult to adapt to large-scale production, resulting in insufficient stability of water control and regulatory lag [12]. Mechanistic models, on the other hand, struggle to capture the complex reactions and nonlinear relationships within the coagulation process, thereby limiting their practical applicability [13]. Alternatively, data-driven models utilizing machine learning can construct black-box models for nonlinear, multivariate dynamic systems by learning from limited datasets [14]. However, current models have limitations in simulating and predicting anticoagulant dosing, particularly in terms of generalizability and convergence [15]. Among the existing methods for coagulation dosing prediction, DL-MV relies on parameter tuning for high-precision prediction [16]; PSO-SVR optimization requires manually preset parameters [17]; and GA-RF architectures are hardened and

2.2. Học máy tự động (AutoML)

Dự đoán có thể diễn giải về liều lượng chất keo tụ trong nhà máy xử lý nước cấp dựa trên học máy tự động và phương pháp SHAP

1. Giới thiệu

2. Vật liệu và phương pháp

2.1. Thu thập và tiền xử lý dữ liệu

Khi thu thập dữ liệu, lỗi đo đạc và sự cố hệ thống chậm tạo ra các giá trị khuyết và nhiễu trong tập dữ liệu thô. Do đó, tiền xử lý là bước thiết yếu để nâng cao độ tin cậy dữ liệu, hiệu suất mô hình và chất lượng ra quyết định [38]. Trong nghiên cứu này, mô hình ước tính K láng giềng gần nhất (KNN, k = 4) được dùng để xử lý dữ liệu khuyết. Mô hình này cân bằng giữa độ ổn định nội suy và tính đại diện toàn cục. Phương pháp giúp bảo toàn dữ liệu tốt hơn cho các thông số quan trọng bị khuyết không ngẫu nhiên (ví dụ: nhiệt độ nước, nitơ amoniac). Hiệu quả nội suy trong khoảng k = 1–10 được khảo sát bằng tìm kiếm lưới. Sai số toàn phương trung bình (MSE) được sử dụng làm tiêu chí đánh giá. Kết quả cho thấy MSE trên tập kiểm tra đạt giá trị nhỏ nhất tại k = 4. Ngoài ra, nhiễu và sai số ngẫu nhiên được giảm thiểu bằng bộ lọc trung bình trượt với kích thước cửa sổ n = 5. Kích thước này được xác định dựa trên phân tích hàm tự tương quan (ACF). Giá trị này ghi nhận hiệu quả các biến động ngắn hạn của thông số chất lượng nước. Đồng thời, nó tránh hiện tượng làm mịn quá mức gây suy giảm đỉnh [39].

Để khảo sát tương quan giữa các đặc trưng khác nhau, chúng tôi sử dụng phương pháp hệ số tương quan thứ hạng Spearman (rs). Các đặc trưng gồm WTR, chỉ số chất lượng nước thô, chỉ số chất lượng nước sau xử lý và liều lượng chất keo tụ. Phương pháp này được biết đến với độ nhạy thấp đối với các mối quan hệ phi tuyến và giá trị ngoại lai. Chúng tôi loại bỏ các đặc trưng có tương quan thấp để giảm bớt đặc trưng dư thừa. Bước này giúp giảm số chiều đầu vào của mô hình, qua đó nâng cao cả hiệu quả vận hành lẫn khả năng tổng quát hóa. Sơ đồ quy trình minh họa phương pháp xây dựng mô hình được trình bày trong Fig. 1.

2.3. Phương pháp diễn giải SHAP

2.4. Đánh giá mô hình

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have high computational requirements [18]. These methods generally have two limitations: first, they rely on manual intervention, making it difficult to automate the adjustment of complex nonlinear water quality relationships; second, they adhere to a single optimization paradigm and lack dynamic synergistic mechanisms, resulting in limited predictive efficiency and generalizability.

Automated Machine Learning (AutoML) builds efficient ML pipelines by automating the entire process [19], and its distributed frameworks (e. g., TPOT) enable fully automated model filtering to reduce development costs and labor dependency [20]. Studies have shown that AutoML excels in hyperparameter optimization and complex constraint solving [21–24]. TPOT improves accuracy by 15–20 % without feature reduction for 50-dimensional water quality data [25]. The H2O framework accurately predicts biological nutrient removal in wastewater [26], and TPOT outperforms traditional ML by 1.4 % in the lake water quality classification task [27]. Nevertheless, its exploration and application in coagulation agent dosing prediction have not yet been extensively pursued.

While some studies have used advanced ML techniques, their black-box nature limits interpretability and hinders practical application in water treatment plants, where operational transparency is critical. The SHapley Additive exPlanations (SHAP) method proposed by Lundberg and Lee [28], which significantly improves model transparency, credibility, and decision making by quantifying the marginal contribution of features to model predictions and identifying key influences and their interaction effects through global and local analysis [29]. The SHAP method has been applied in many fields [30–34] and has been used with high accuracy for surface water quality prediction [35–37]. However, this method has not been applied to coagulant dosage prediction.

Based on existing research, this study applies the interpretable AutoML framework to coagulation dose prediction for the first time, constructs the prediction system through automated model selection and hyper-parameter optimization, and integrates the SHAP method to achieve the global feature importance and local decision interpretation of the black-box model. By using real-time data and historical information, it ensures that the coagulation process can be accurately controlled while maintaining treated water quality standards, facilitating future technological advances and upgrades in the water treatment industry.

2. Materials and methods

2.1. Data acquisition and preprocessing

The data utilized in this study were sourced from a water treatment plant located in Tianjin, China. This plant receives water from two primary sources: the Luanhe River (for December, January and February annually), diverted from Hebei Province, and the Yangtze River (from March to November annually), originating from the Danjiangkou Reservoir in Hubei Province. The water treatment process involves pre-treatment, enhanced conventional treatment, and a combination of ultraviolet and chlorine disinfection. The daily treatment capacity of the water plant is 17.5 wt/d, and the hydraulic retention time is relatively stable at 25.56 min. The plant uses two types of coagulants: ferric chloride and PACl, which are dosed during the pretreatment and blending stages, respectively. Water quality parameters and coagulant dosing data were collected daily from October 2022 to May 2024, totaling 1339 data points. These data were randomly divided into a training set (1071 entries) and a test set (268 entries) in an 8: 2 ratio. The feature variables consist of inflow rate (WTR) and water quality indicators such as temperature (T), pH, turbidity (NTU), and electrical conductivity (EC), and chemical oxygen demand determined by potassium permanganate (COD_{Mn}) for both raw water (RW) and treated water (TW), with ammonia nitrogen (NH₃–N) being considered only for raw water. Based on practical plant operations and professional evaluations, the dosing amounts of the two coagulants are comparable, with a

maximum discrepancy not exceeding 5 mg/L. Therefore, PACl was used as the only target variable for accurate simulation of coagulant dosing in this study.

During data collection, the occurrence of detection errors and dosing system malfunctions resulted in the raw data set containing problems such as missing values and noise. Therefore, preprocessing is essential to improve data reliability, model performance, and decision quality [38]. In this study, the K-nearest neighbor (KNN, k = 4) estimation model was used to deal with missing data, which balances interpolation stability and global representativeness, and provides better data retention for non-randomly missing key parameters (e.g., water temperature, ammonia nitrogen). The interpolation effect in the range of k = 1–10 was searched through the grid and the mean square error (MSE) was used as the evaluation criterion, and the results showed that the MSE was minimized on the test set at k = 4. In addition, noise and random errors were mitigated using a moving average filter with a window size of $n = 5$, determined based on an autocorrelation function (ACF) analysis, a value that effectively captures short-term fluctuations in water quality parameters while avoiding excessive smoothing that leads to peak attenuation [39].

To investigate the correlations between various features such as WTR, raw water quality indicators, treated water quality indicators, and coagulant dosage, we used the Spearman rank correlation coefficient (rs) method, which is known for its low sensitivity to nonlinear relationships and outliers. By excluding features with low correlations, we reduced redundant features to reduce the input dimensionality of the model, thereby improving both the operational efficiency and generalization capabilities of the model. A flowchart illustrating the modeling methodology is shown in Fig. 1.

2.2. Automated machine learning

AutoML is a technique for automating the machine learning process through algorithms, including feature engineering, algorithm selection, and hyperparameter optimization [40]. Its primary goal is to reduce the need for human intervention while improving model performance. In this study, the Tree-based Pipeline Optimization Tool (TPOT), an AutoML tool based on genetic programming, is used to automate model construction. TPOT is an open source Python-based AutoML library (version 0.11.7) with underlying dependencies on Scikit-Learn (version 1.0.2), DEAP (Distributed Evolutionary Algorithms Framework), and extensions such as XGBoost [27]. The model runs in a Python 3.8 environment, and the full configuration code for TPOT is open source.

In this study, the TPOT framework is adopted to automate the design of machine learning pipelines, whose core mechanisms are: iterative optimization based on tree-structured coding of genetic programming, integration of Scikit-Learn multi-algorithms with predefined hyperparameter space to balance the search efficiency; and screening of the Pareto-optimal pipeline by repeating the 10-fold cross-validation (run independently for 3 times) to ensure the model generalization performance [41].

2.3. SHAP interpretability method

The SHAP methodology encompasses Kernel SHAP, Deep SHAP, and Tree SHAP [42]. This study used the Tree SHAP package in Python 3.9 to analyze the importance, dependencies, and interactions of features. The modules and functions used in Python are listed in Table 1. The average absolute SHAP values were used to evaluate the importance of different factors in determining the dosage amount. By summing the average absolute SHAP values for all relevant environmental parameters, their combined impact on dosing strategies was quantified. Dependency plots of SHAP values revealed interactions between features and their collective contribution to the prediction of dosing strategies for the coagulation process [43]. Two typical samples were selected to generate local interpretability plots, waterfall plots, and decision plots in order to

3. Kết quả và thảo luận

3.1. Phân tích dữ liệu chất lượng nước và lựa chọn đặc trưng

Fig. 2 minh họa phân phối biểu đồ tần số và các chỉ số thống kê của từng đặc trưng. Các chỉ số này gồm giá trị nhỏ nhất (MIN), lớn nhất (MAX), trung bình, độ lệch chuẩn (SD) và hệ số biến thiên (CV). CV là thước đo định lượng độ biến động trong các tập dữ liệu [47]. Giá trị CV càng cao cho thấy độ biến động dữ liệu càng lớn. Đáng chú ý, nước thô (RW) có sự khác biệt lớn ở T (SD = 9.37, CV = 62 %), NH₃-N (SD = 0.07, CV = 59 %) và NTU (SD = 2.58, CV = 57 %). Nước sau xử lý (TW) cũng ghi nhận biến động ở chỉ số T (SD = 8.77, CV = 54 %). Hầu hết các đặc trưng khác tương đối ổn định. Dao động pH rất nhỏ trong toàn bộ quy trình xử lý nước. Giá trị CV chỉ đạt 3 % đối với pH nước thô (RW) và 2 % đối với pH nước sau xử lý (TW).

Table 1
Các mô-đun Python và chức năng tương ứng được sử dụng.

Ma trận tương quan thứ hạng Spearman trong Fig. 3 minh họa mối quan hệ giữa các thông số đầu vào và biến kết quả. Liều lượng chất keo tụ PACl tương quan mạnh với T-RW (rs = −0.71), NH₃-N-RW (rs = 0.87) và pH-RW (rs = −0.9). Tương quan mạnh này cũng được ghi nhận ở T-RW (rs = −0.67) và WTR (rs = −0.68). Tương quan trung bình xuất hiện ở NTU-RW (rs = 0.45), EC-RW (rs = 0.56), pH-RW (rs = −0.33) và CODMn-RW (rs = 0.56). Mức tương quan trung bình cũng được tìm thấy ở EC-TW (rs = 0.43) và NTU-TW (rs = −0.23). CODMn-TW (rs = 0.011) thể hiện tương quan yếu với liều lượng PACl. Đáng chú ý, EC-RW và EC-TW có tương quan tuyến tính rất cao (rs = 0.98). CODMn-RW và EC-RW cũng tương quan tuyến tính cao với T-TW (lần lượt có rs = 0.91 và −0.95). Do đó, để tránh dư thừa, mô hình chỉ giữ lại một đặc trưng từ mỗi cặp có tương quan cao, bên cạnh EC-RW và CODMn-RW. Tập đặc trưng cuối cùng để huấn luyện mô hình gồm T-RW, pH-RW, NH₃-N-RW, NTU-RW, EC-RW, CODMn-RW, pH-TW, NTU-TW và WTR. Biểu mục tiêu của mô hình là liều lượng PACl. Mô hình này nắm bắt các tương tác phức tạp giữa chất lượng nước thô (RW), liều lượng châm và chất lượng nước sau xử lý (TW).

Fig. 2. Phân phối biểu đồ tần số của các đặc trưng và biến mục tiêu (RW: nước thô, TW: nước sau xử lý, MIN: giá trị nhỏ nhất, MAX: giá trị lớn nhất, SD: độ lệch chuẩn, CV: hệ số biến thiên). Mỗi đồ thị còn thể hiện phân phối của các điểm mẫu trong một biến cụ thể. Trục tung biểu thị tần suất xuất hiện của từng đặc trưng trong tập dữ liệu. Trục hoành biểu diễn các khoảng mẫu. Đường màu tím biểu thị đường cong phân phối tần số được làm khớp, chỉ ra phạm vi sử dụng biến trong tập dữ liệu). (Để hiểu rõ các chú dẫn về màu sắc trong hình vẽ này, người đọc vui lòng tham khảo phiên bản trực tuyến của bài báo).

Fig. 4 trình bày kết quả huấn luyện và kiểm tra của mô hình RF được phát triển bằng AutoML để dự đoán liều lượng chất keo tụ. Như thể hiện trong Fig. 4a và b, mô hình thể hiện hiệu suất ổn định và độ chính xác cao trên cả tập huấn luyện và kiểm tra. Hàm lượng chất rắn lơ lửng, hạt keo và các chất ô nhiễm khác trong nước thô ở mức thấp giúp nước có chất lượng cao. Ảnh hưởng tối thiểu của pH-RW và NTU-RW lên quá trình keo tụ khiến sai số trên tập kiểm tra chủ yếu dưới 20 mg/L. Biểu đồ mật độ phân tán trong Fig. 4c và d cho thấy liều lượng PACl thực tế và dự đoán chủ yếu nằm dọc đường chéo. Điều này cho thấy độ chệch tiềm ẩn của mô hình ở mức tối thiểu.

Table 2
Kết quả tối ưu hóa siêu tham số của các mô hình.

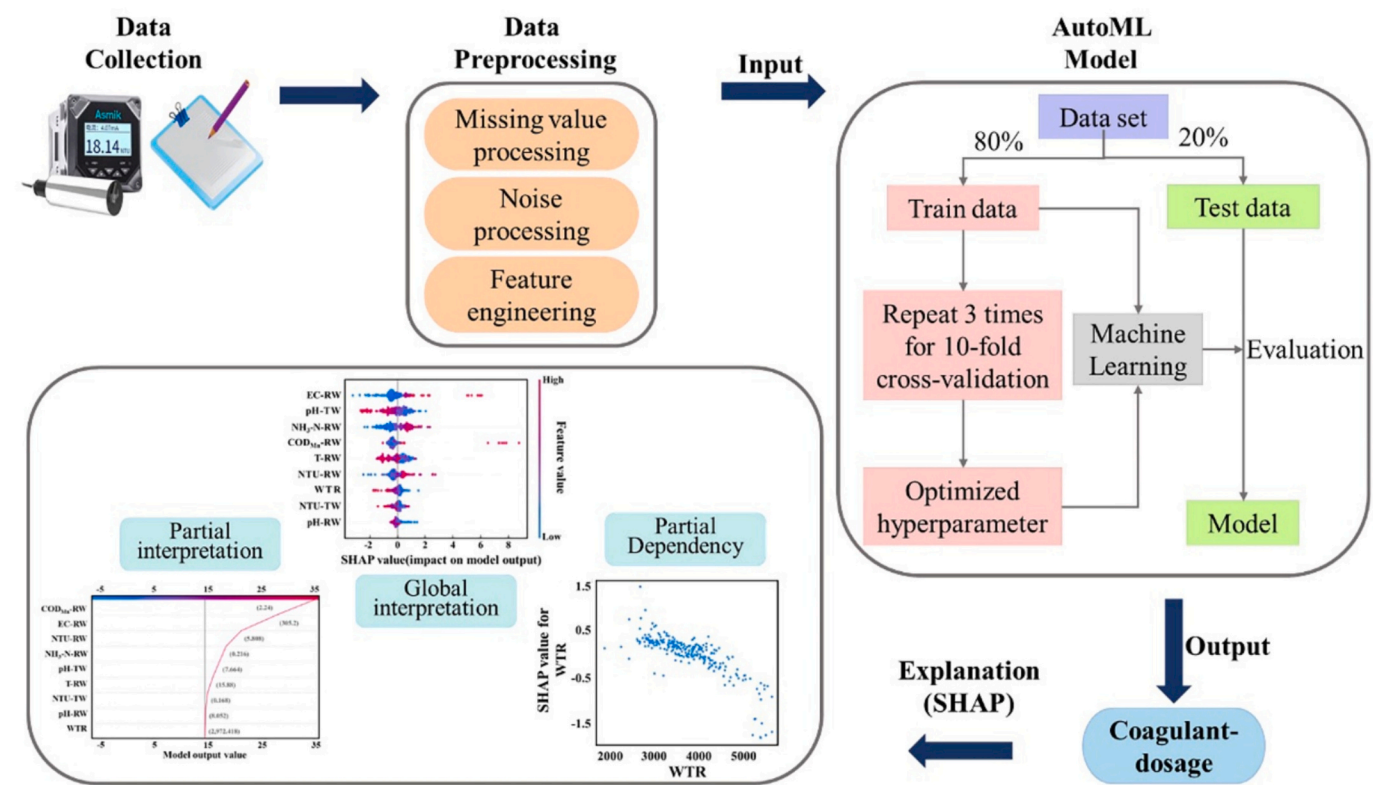


Fig. 1. A flowchart illustrates the modeling methodology.

Table 1	
Python modules and their corresponding functions employed.	
Module	Function
Sklearn 1.5.0	Data processing, data set division, model evaluation, cross validation
Scipy 1.13.1	Spearman analysis
Seaborn 0.13.2	Statistical data visualization
Tpot 0.12 2	TPOT Regressor model establishment
Shap 0.45.1	Model interpretation, data visualization
Matplotlib 3.9.0	Graphical drawing

illustrate feature importance and decision-making mechanisms within the samples.

2.4. Model evaluation

Root mean square error (RMSE), mean absolute error (MAE), and coefficient of determination (R^2) were used as evaluation metrics to assess the reliability and accuracy of the predictive models [44]. RMSE assesses the degree of discrepancy between predicted and actual values within a sample set [45]. It has an increased sensitivity to outliers and acts as a metric of model accuracy by calculating the average size of prediction errors. MAE reflects the average discrepancy between predicted and actual values, with values closer to zero indicating superior model performance [10]. R^2 quantifies the fit of model predictions to actual data, assessing the robustness and reliability of the model. A value of R^2 approaching one indicates a higher correlation between the dependent and independent variables, which indicates a better model fit and consequently smaller prediction errors [46].

3. Results and discussion

3.1. Analysis of water quality data and feature selection

Fig. 2 illustrates the histogram distributions and statistical indicators, including minimum (MIN), maximum (MAX), mean, standard deviation (SD), and coefficient of variation (CV) for each feature. CV serves as a metric to quantify the variability in datasets [47], with higher CV values indicating greater data variability. Notably, significant differences are observed in T (SD = 9.37, CV = 62 %), $\text{NH}_3\text{-N}$ (SD = 0.07, CV = 59 %), NTU (SD = 2.58, CV = 57 %) for raw water, and T for treated water (SD = 8.77, CV = 54 %). Most other features remain relatively stable, with the pH fluctuation was minimal throughout the entire water treatment process, with a CV of only 3 % for the raw water pH and a CV of merely 2 % for the treated water pH.

The quality of raw water has a direct impact on the safety and reliability of the final water supply, as well as coagulant requirements. Raw water quality indicators at this plant, such as COD_{Mn} and ammonia nitrogen, exhibit distinct seasonal trends. During summer, elevated temperatures and increased rainfall stimulate algae growth in water sources, which raises the COD_{Mn} in water bodies and subsequently affects raw water quality. Additionally, rising temperatures reduce water viscosity, promoting ion movement in solutions and leading to decreased coagulant requirements. In contrast, during winter, low temperatures decrease the solubility and diffusivity of coagulants, resulting in a deceleration of chemical reactions. Consequently, higher coagulant dosages become necessary to ensure the effectiveness of water treatment. Accurate calculation of coagulant dosage based on real-time raw water quality data and historical fluctuation information is crucial for maintaining process efficiency, stable water quality output, and prudent chemical use.

The Spearman rank correlation matrix in Fig. 3 illustrates the relationships between the input parameters and the outcome variable. Strong correlations with PACl coagulant dosage are observed for T-RW

Fig. 7 minh họa các giá trị SHAP cho từng biến riêng lẻ, thể hiện hiệu ứng biến của từng đặc trưng lên dự đoán của mô hình.

So với kinh nghiệm thủ công, phương pháp này dự đoán chính xác hơn liều lượng tối ưu cần thiết. Mô hình giúp nước luôn đạt chuẩn sau xử lý dưới các điều kiện nước thô biến động. Đồng thời, phương pháp này tiết kiệm 10.25 % liều lượng chất keo tụ mỗi năm. Đánh giá thủ công thường dẫn đến chậm hóa chất quá mức khi chất lượng nước thô vượt quá một ngưỡng nhất định. Ngược lại, dự đoán từ mô hình cho phép chậm hóa chất chính xác mà vẫn duy trì tiêu chuẩn nước sau xử lý. Quá trình này giúp giảm chi phí hóa chất và đảm bảo chất lượng nước sau xử lý ổn định hơn.

Table 3
So sánh hiệu suất của các mô hình học máy khác nhau.

3.3. Diễn giải mô hình bằng SHAP

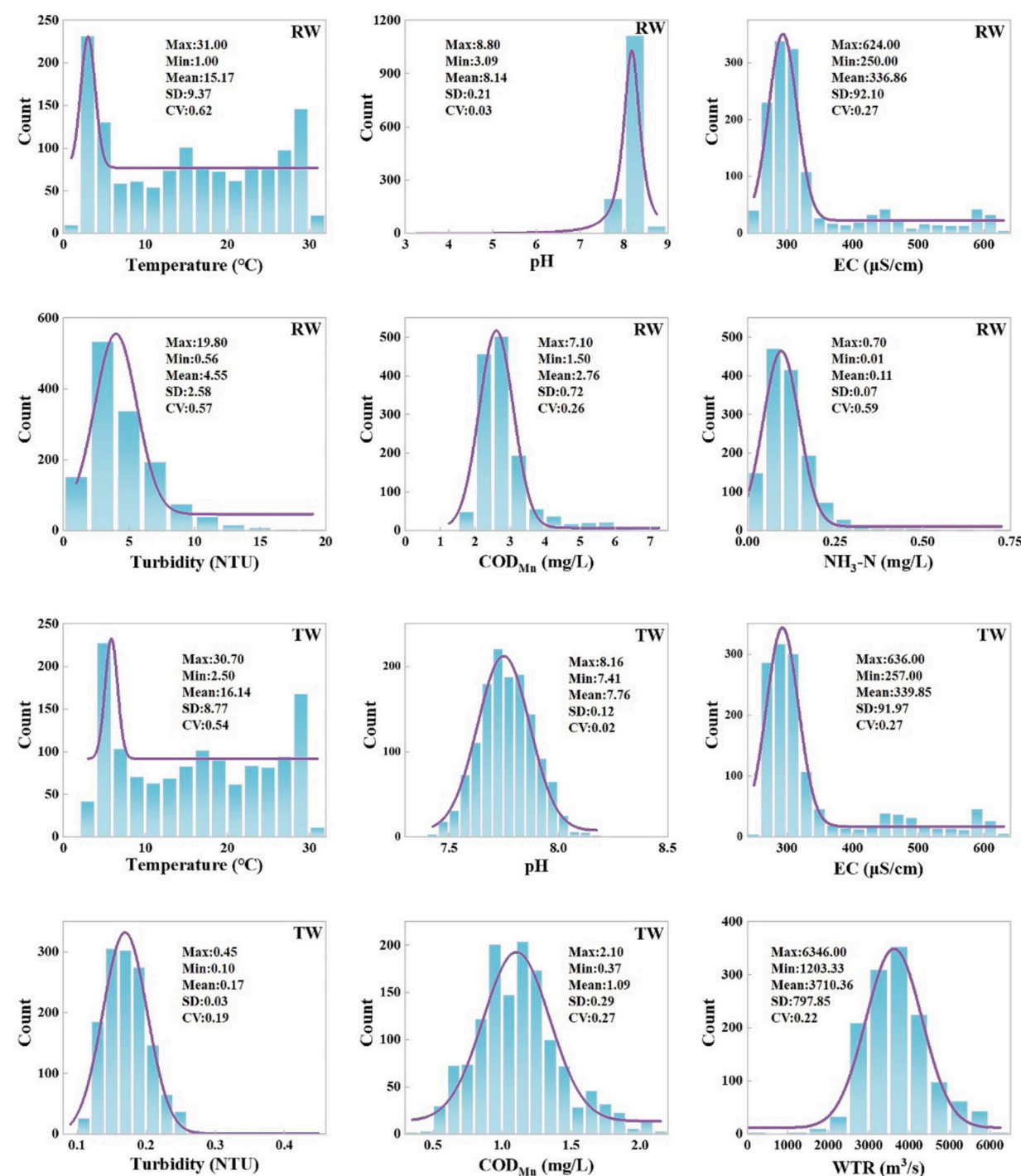


Fig. 2. Histogram distributions of various features and target variables (RW: raw water, TW: treated water, MIN: minimum, MAX: maximum, SD: standard deviation, CV: coefficient of variation. Each subplot represents the distribution of sample points within a specific variable. The vertical axis reveals the occurrence frequency of each feature within the dataset, while the horizontal axis depicts the sample intervals. The purple line represents the fitted frequency distribution curve, which indicates the range of variable utilization within the dataset). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

($r_s = -0.71$), $\text{NH}_3\text{-N-RW}$ ($r_s = 0.87$), pH-RW ($r_s = -0.9$), T-RW ($r_s = -0.67$), and WTR ($r_s = -0.68$). Moderate correlations are found for NTU-RW ($r_s = 0.45$), EC-RW ($r_s = 0.56$), pH-RW ($r_s = -0.33$), and $\text{COD}_{\text{Mn-RW}}$ ($r_s = 0.56$), as well as for EC-TW ($r_s = 0.43$) and NTU-TW ($r_s = -0.23$). $\text{COD}_{\text{Mn-TW}}$ ($r_s = 0.011$) shows a weak correlation with PACl dosage. Notably, EC-RW and EC-TW are highly linearly correlated ($r_s = 0.98$), and $\text{COD}_{\text{Mn-RW}}$ and EC-RW are also highly linearly correlated with T-TW ($r_s = 0.91$ and -0.95 , respectively). Therefore, to avoid

redundancy, only one feature from each highly correlated pair is retained, in addition to EC-RW and $\text{COD}_{\text{Mn-RW}}$. The final feature set for model training includes T-RW , pH-RW , $\text{NH}_3\text{-N-RW}$, NTU-RW , EC-RW , $\text{COD}_{\text{Mn-RW}}$, pH-TW , NTU-TW , and WTR , with PACl dosage as the target variable. This model captures the complex interactions between raw water quality, dosage, and treated water quality.

Ngoài ra, liều lượng chất keo tụ cần thiết giảm khi pH-RW tăng trong khoảng từ 8.0 đến 8.4. Trong môi trường kiềm yếu, PACl và hạt ô nhiễm mang điện tích trái dấu [55]. Đặc tính này giúp PACl thể hiện hiệu quả keo tụ vượt trội [56]. Nhiệt độ nhìn chung có tác động âm lên việc châm chất keo tụ. Quá trình thủy phân PACl là phản ứng thu nhiệt. Độ nhớt tăng ở nhiệt độ thấp cản trở chuyển động nhiệt của hạt lơ lửng, gây bất lợi cho sự kết tụ keo ổn định [57]. Do đó, cần tăng liều lượng PACl để đạt hiệu quả keo tụ tối ưu [58].

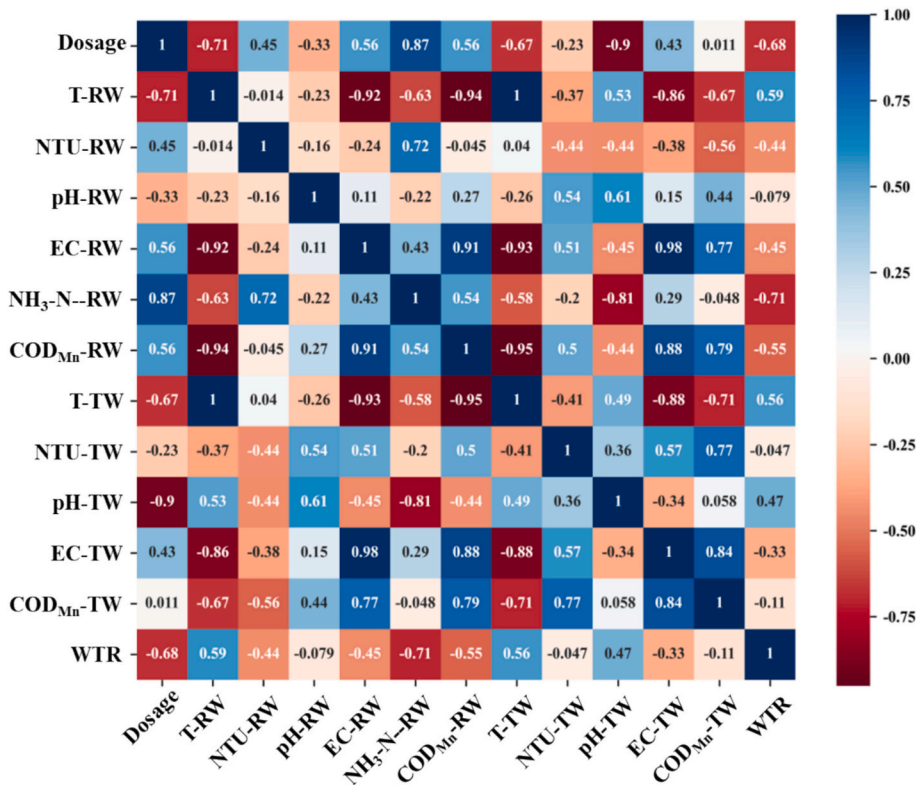


Fig. 3. Spearman rank correlation coefficient matrix between all input parameters and the target variable. Symbols with the suffix “RW” represent raw water quality indicators, while those with “TW” are for treated water.

3.2. Model comparison and optimization

In this study, alongside TPOT AutoML, a range of typical algorithms were chosen to develop a prediction model of PACl dosage for comparison, including Decision Tree (DT), Support Vector Regression (SVR), K-Nearest Neighbors (KNN), Linear Regression (LR), and Gradient Boosting Tree (GBT). Table 2 lists the hyperparameters of the models and Table 3 lists the performance metrics for each model. The optimal algorithm selected by AutoML was Random Forest (RF), which provided more robust and accurate predictions compared to the other algorithms. The R² values of the RF model for the training and testing sets reached 1.00 and 0.96, respectively, with the corresponding lowest RMSE: 0.95 ± 0.11 mg/L and MAE: 0.47 ± 0.83 mg/L. Kim et al. [48] predicted PAC dose using a convolutional neural network and gated recurrent units, and the model R² exceeded 0.8 for both the training and test sets. This may be due to the fact that the model is typically used to extract spatiotemporal features, and the main drivers of PAC dose prediction are static water quality parameters (e.g., pH, turbidity, temperature, etc.) rather than time series or spatial patterns, which can introduce unnecessary complexity and lead to information redundancy. Lin et al. [39] used deep learning to predict PAC and its sulfate with a model RMSE of 3.55 and an R² of 0.94, while the error was even smaller when predicting

Table 2
Hyperparameter optimization results of the models.

Type of model	Hyperparameter setting
DT	max_depth = 8, min_samples_split = 10, min_samples_leaf = 5, max_features = 0.5, random_state = 42
SVR	c = 1, epsilon = 10, gamma = scale
KNN	n_neighbors = 15, p = 1, metric = minkowski
LR	fit_intercept = True, normalize = True, positive = False
GBT	n_estimators = 150, learning_rate = 0.1, max_depth = 5, min_samples_split = 8, subsample = 0.8, max_features = sqrt

Table 3
Performance comparison of different machine learning models.

Type of model	Training			Testing		
	RMSE	MAE	R ²	RMSE	MAE	R ²
AutoML: RF	0.07	0.02	1.00	0.89	0.47	0.96
GBT	1.06	0.74	0.96	1.41	0.97	0.91
LR	1.68	1.21	0.89	1.81	1.34	0.84
DT	0.00	0.00	1.00	2.07	0.66	0.80
KNN	2.90	1.91	0.68	3.46	2.55	0.43
SVR	4.81	2.85	0.13	4.45	2.89	0.06

coagulants using AutoML. Genetic programming for TPOT often favors integrated methods, which are inherently better for small- to medium-sized data with better bias-variance balance. The worst performance was observed for SVR, with R² values of only 0.13 and 0.06 for the training and test sets, respectively. This is likely due to the high sensitivity of SVR to hyperparameter selection and the lack of a precise solution in defining the configuration parameters [49]. DT achieved an R² of 1.00 on the training set, but performed poorly on the testing set because it tends to build complex models on the training data that capture noise and details rather than the true data distribution patterns, resulting in poor predictive performance on new data.

Fig. 4 shows the training and testing results of the RF model developed using AutoML for predicting coagulant dosage. As shown in Fig. 4a and b, the model exhibits robust performance and high accuracy on both the training and testing sets. The low concentrations of suspended solids, colloids, and other contaminants in the raw water result in high quality water, and the minimal influence of pH-RW and NTU-RW on the coagulation process means that the errors observed in the testing dataset are predominantly below 20 mg/L. From the scatter density plots in Fig. 4c and d, it can be observed that the actual and predicted PACl dosages on the training and testing set are mostly aligned along the diagonal, indicating a minimal potential bias in the model.

Fig. 8 thể hiện biểu đồ thác nước và biểu đồ quyết định cho hai mẫu cục bộ cụ thể bằng phương pháp SHAP. Fig. 8a và b thể hiện mẫu có liều lượng chất keo tụ cao hơn giá trị trung bình (14.5 mg/L). Mẫu này được thu thập vào tháng 2 với nguồn nước thô lấy từ sông Loan Hà và T-RW đạt 3.8 °C. Fig. 8a cho thấy liều lượng chất keo tụ dự đoán của mẫu này (34.9 mg/L) cao hơn mức kỳ vọng trung bình (14.5 mg/L). Nguyên nhân chính do CODMn-RW, tiếp theo là EC-RW, NTU-RW và NH3-N-RW, trong khi đóng góp của pH-RW và WTR là không đáng kể. Fig. 8b thể hiện đóng góp của từng đặc trưng khi vượt mức dự đoán trung bình (14.5 mg/L). Các đóng góp này tạo ra giá trị đầu ra cuối cùng là 34.9 mg/L cho mẫu. Đầu tiên, khi CODMn-RW đạt 5.46 mg/L và vượt ngưỡng 4 mg/L, giá trị SHAP tăng lên đáng kể. Ngoài ra, với EC-RW ở mức 600.6 μS/cm, NH3-N-RW ở mức 0.36 mg/L và NTU-RW đạt 7.19 NTU, các giá trị SHAP đều duy trì trên 0. Dựa trên đánh giá chuyên môn về tác động tổng thể của chất lượng nước thô, liều lượng PACl sau đó được tăng lên từ mức trung bình.

Tuyên bố đóng góp của tác giả (CRediT)

Tuyên bố về xung đột lợi ích

4. Kết luận

(2) Diễn giải mô hình dựa trên giá trị SHAP làm rõ ảnh hưởng của từng chỉ số chất lượng nước đến kết quả dự đoán. Cụ thể, khi NTU-RW thấp và tương đối ổn định, độ dẫn điện nước thô tác động mạnh nhất đến việc châm chất keo tụ. Theo sau là các yếu tố NH3-N-RW, CODMn-RW, T-RW và WTR.

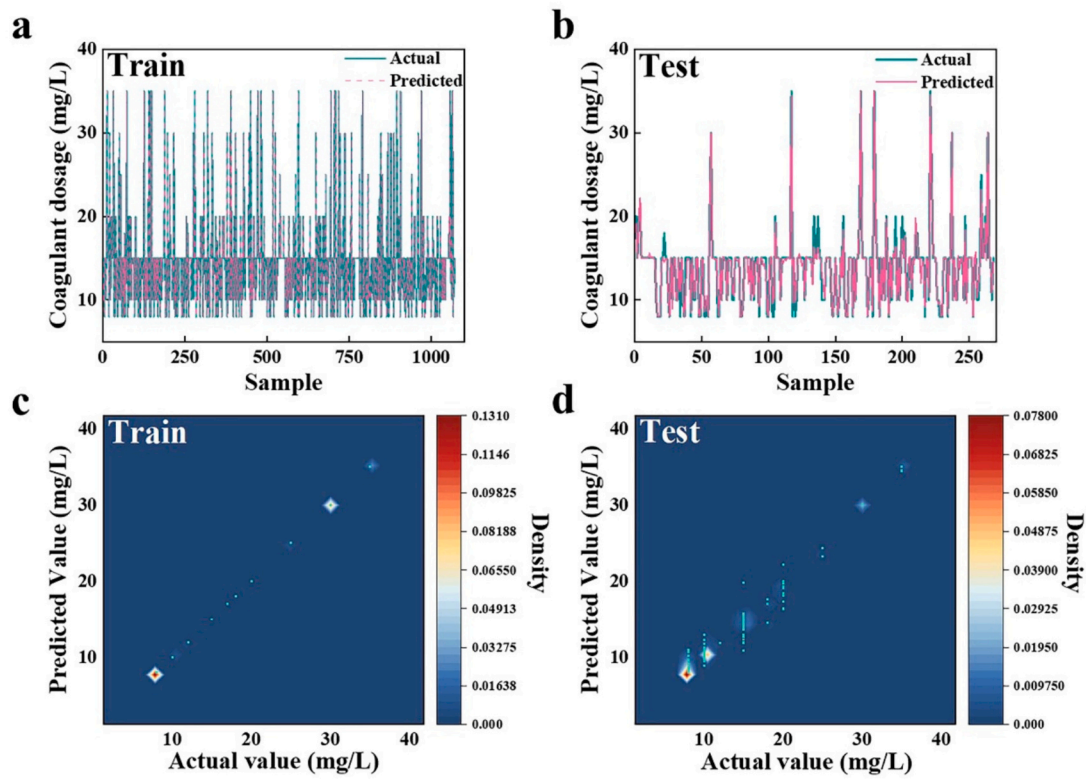


Fig. 4. Fitting results and scatter density plots for the AutoML: RF model on training and testing sets. The Fig includes model performance plot for samples on the training set (a) and the testing set (b), and scatter plot of samples on the training set (c) and the testing set (d).

Fig. 5 shows the optimized dosage of PACl for two different water sources: Yangtze River water and Luanhe River water. The gray areas in the Fig represent the reduction in PACl dosage before and after optimization, highlighting the savings achieved. The optimized PACl dosage was predicted using the optimal RF model, which uses actual raw water quality data with specific targets for treated water quality. For Yangtze River water, the targets were set at NTU-TW: 0.2 NTU, COD_{Mn}-TW: 1.2 mg/L, pH-TW: 8, and NH₃-N-TW: 0.01 mg/L. For Luanhe River water, the targets were: NTU-TW: 0.3 NTU, COD_{Mn}-TW: 1.5 mg/L, pH-TW: 8, and the NH₃-N-TW: 0.01 mg/L. In spring and summer, the temperature of Yangtze River water is high, PACl hydrolysis is fast, and algae in the water source are negatively charged and proliferate, so more PACl is needed for charge neutralization. By precisely adjusting the dosage, the empirical redundancy at high temperature can be avoided, the dosage of chemicals can be reduced by 222 kg/d, saving 180.7 RMB/day, and the consumption of PACl can be reduced by 11 % per year. Luanhe water

temperature in winter is low, Al³⁺ hydrolysis is incomplete, floc formation is slow, and the formation of virtual body of small particle size, low density, but also may be affected by the upstream agricultural pollution, these need to compensate for the additional amount of PACl, through the optimization of PACl dosage prediction, it can reduce the compensatory overdosage caused by the low temperature, saving the amount of dosing from 225 kg/d, into 183.2 RMB/day, the amount of medicine can be saved by 8 % per year. The plant uses Yangtze River water as the sole source between March and November each year, with manual PACl dosages ranging from 8 to 30 mg/L. During the low temperature months from December to February each year, the manual PACl dosages for Luanhe River water are higher, ranging from 10 to 35 mg/L, which potentially allows for higher savings dosages, but at a lower percentage. As a result, according to the weighted average, the model has the potential to save 10.25 % of the annual dosage for the plant.

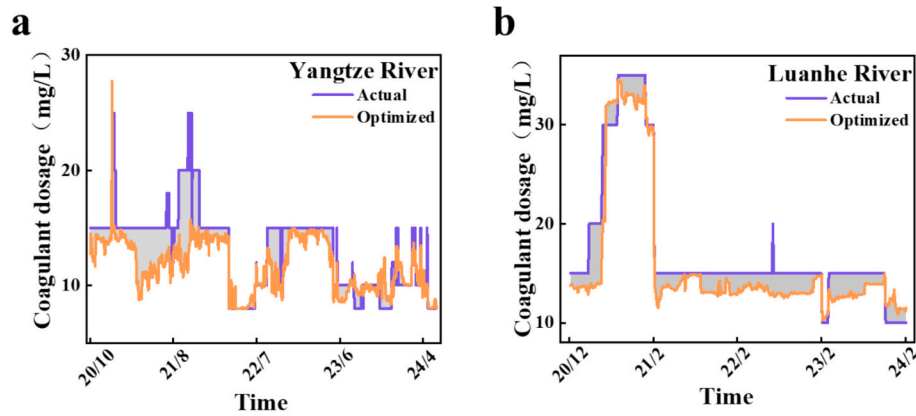


Fig. 5. Optimization results of PACl dosage using the AutoML: RF model for the Yangtze River source (a) and the Luanhe River source (b).

Tính khả dụng của dữ liệu

Lời cảm ơn

Nghiên cứu này nhận tài trợ từ Chương trình Nghiên cứu và Phát triển Trọng điểm Quốc gia (2022YFC3203803). Nghiên cứu cũng được hỗ trợ bởi Sở Khoa học và Công nghệ tỉnh Phúc Kiến (2022J01522). Chúng tôi chân thành cảm ơn Công ty Công nghiệp Nước TEDA Thiên Tân đã cung cấp dữ liệu thô.

Dữ liệu sẽ được cung cấp khi có yêu cầu.

This method more accurately predicts the optimal dosage required to consistently meet treated water standards under varying raw water conditions than manual empirical judgment and saves 10.25 % of coagulant dosage per year. Manual judgment often results in overdosing when raw water quality exceeds a certain threshold. Conversely, model prediction enables accurate dosing while maintaining treated water standards, reducing reagent costs and ensuring more stable treated water quality.

3.3. SHAP interpretation

Fig. 6 presents the SHAP feature importance and summary plots for the model evaluated on the testing set. Feature importance ranking from highest to lowest is as follows: EC-RW, pH-TW, $\text{NH}_3\text{-N-RW}$, $\text{COD}_{\text{Mn-RW}}$, T-RW, NTU-RW, WTR, NTU-TW, and pH-RW, as shown in Fig. 6(a). Notably, the model in this study was constructed using monitoring data from the water treatment plant, adhering to the principle of continuously adjusting dosage based on raw water quality to ensure stable and compliant treated water quality. Therefore, the relationship between dosage and treated water quality exhibits a direct causal link, despite dosage exceeding a certain threshold may have minimal impact on treated water quality. On the other hand, the relationship between dosage and raw water quality reflects practical expertise, with water plant operators relying on their experience and raw water quality to set chemical dosages that meet treated water standards. Because of the wide variation in raw water quality and the stability of treated water quality, raw water indicators are generally more important than treated water indicators in the model.

Fig. 6 (b) displays the contribution of input features to the prediction results, with features arranged in descending order along the vertical axis. Consistent SHAP values aligned with characteristic values for EC-RW, $\text{NH}_3\text{-N-RW}$, $\text{COD}_{\text{Mn-RW}}$, and NTU-RW suggest a positive influence on coagulant dosing, while the opposite trend for pH-RW, T-RW, and WTR suggests a negative influence on dosing. For example, higher EC-RW corresponds to higher SHAP values, indicating an increase in dosing. Higher EC implies lower water purity, potentially containing higher concentrations of dissolved salts, organics, metal ions, and other contaminants. The cations in PACl can neutralize the surface charges of negatively charged suspended particles or colloidal substances in water, reducing the electrostatic repulsion between particles and promoting aggregation [50]. In contrast, higher WTR values correspond to lower SHAP values, indicating a decrease in dosage.

Fig. 7 illustrates the SHAP values for individual variables, which demonstrates the marginal effects of each feature on model predictions.

For the raw water indicator, conductivity reflects the total concentration of dissolved ions in the water and shows a nearly linear enhancement relationship with dosage when it is between 300 and 550 $\mu\text{S}/\text{cm}$, while below 300 $\mu\text{S}/\text{cm}$ or above 550 $\mu\text{S}/\text{cm}$ this relationship shows a trend of diminishing returns. High conductivity compresses the bilayer and promotes coagulation, but low pH inhibits $\text{Al}(\text{OH})_3$ formation when incremental compensation is required. Ammonia is nitrogen in the form of free ammonia (NH_3) and ammonium ions (NH_4^+) regulated by pH and temperature. $\text{NH}_3\text{-N-RW}$ in the range of 0–0.2 mg/L also exerts a positive influence on coagulant dosage, with the relationship becoming less significant beyond this range. High temperature accelerates the hydrolysis, but low $\text{NH}_3\text{-N}$ can reduce the interference, at this time the PACl dosage can be dosed according to professional judgment.

COD_{Mn} reflects the total amount of organic matter and reducing inorganic matter in the water body, there is no obvious correlation with the dosage, if it is less than 4 mg/L, and more than 4 mg/L threshold, it will compete for the active site with the PACl, and the SHAP value at this time shows a sharp upward trend. Furthermore, NTU-RW generally has a positive promoting effect on dosing, as an appropriate increase in coagulant to promote particle aggregation and settlement, which is in line with the findings of Vunain et al. [51]. However, when NTU-RW is less than 2 NTU, the SHAP value shows an upward trend, as sparse and dispersed colloidal particles in low-turbidity water limit the ability of colloids formed after PACl hydrolysis to capture solid impurity particles. Therefore, the dosage of PACl must be increased to compensate for diffusion limitations and to ensure electroneutralization [52–54]. In high COD_{Mn} and high turbidity water, organic matter can encapsulate colloidal particles and additional amounts of PACl are needed to enhance charge neutralization and break down the organic matter barrier.

In addition, the required coagulant dosage decreases as the pH-RW increases within the range of 8.0 to 8.4, which is attributed to the opposite charges of PACl and contaminant particles in a weakly alkaline environment [55], allowing PACl to exhibit superior coagulation properties [56]. It is likely that temperature typically has a negative effect on dosing. The hydrolysis of PACl is an endothermic reaction, but the increased viscosity at low temperatures impedes the thermal motion of suspended particles in water, which is detrimental to the stability of colloidal aggregation [57]. Therefore, the dosage of PACl should be increased to achieve optimal coagulation efficacy [58].

For treated water indicators, a distinct negative linear correlation is observed between pH-TW and the quantity of coagulant administered, primarily clustering within a narrow pH range of 7.6 to 7.8. This negative linear relationship is primarily due to the fact that the

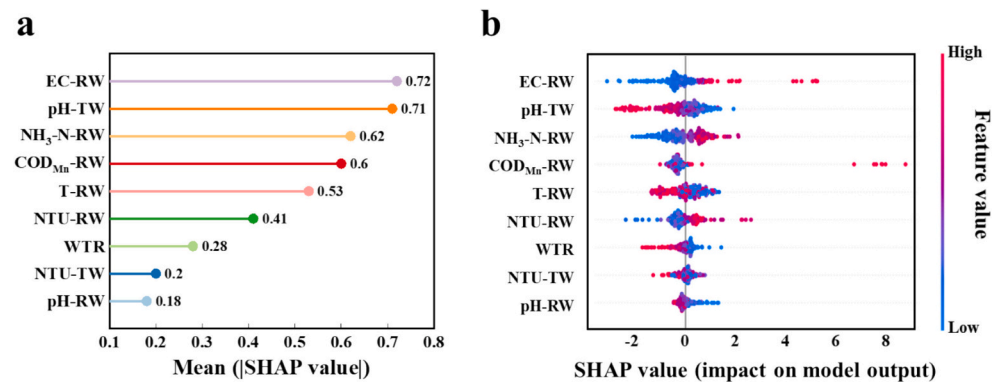


Fig. 6. SHAP feature importance (a) and summary plot (b) for the test set. The vertical axis of plot (a) represents the features in descending order of their average absolute SHAP value on the model output, with larger values indicating a higher contribution of the feature to the prediction, and the horizontal axis indicates the size of the SHAP value, reflecting the average degree of influence of the feature on the model output. Fig (b) each point represents the SHAP value of a sample, the position of the horizontal axis shows the size of the value, the colour shades reflect the original value of the feature is high or low (red is high value, blue is low value), the vertical axis represents the order by the importance of the feature, in line with Fig (a), the superimposed trend represents the relationship between the feature's value high or low and the direction of the SHAP value that can be observed from the distribution of points. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

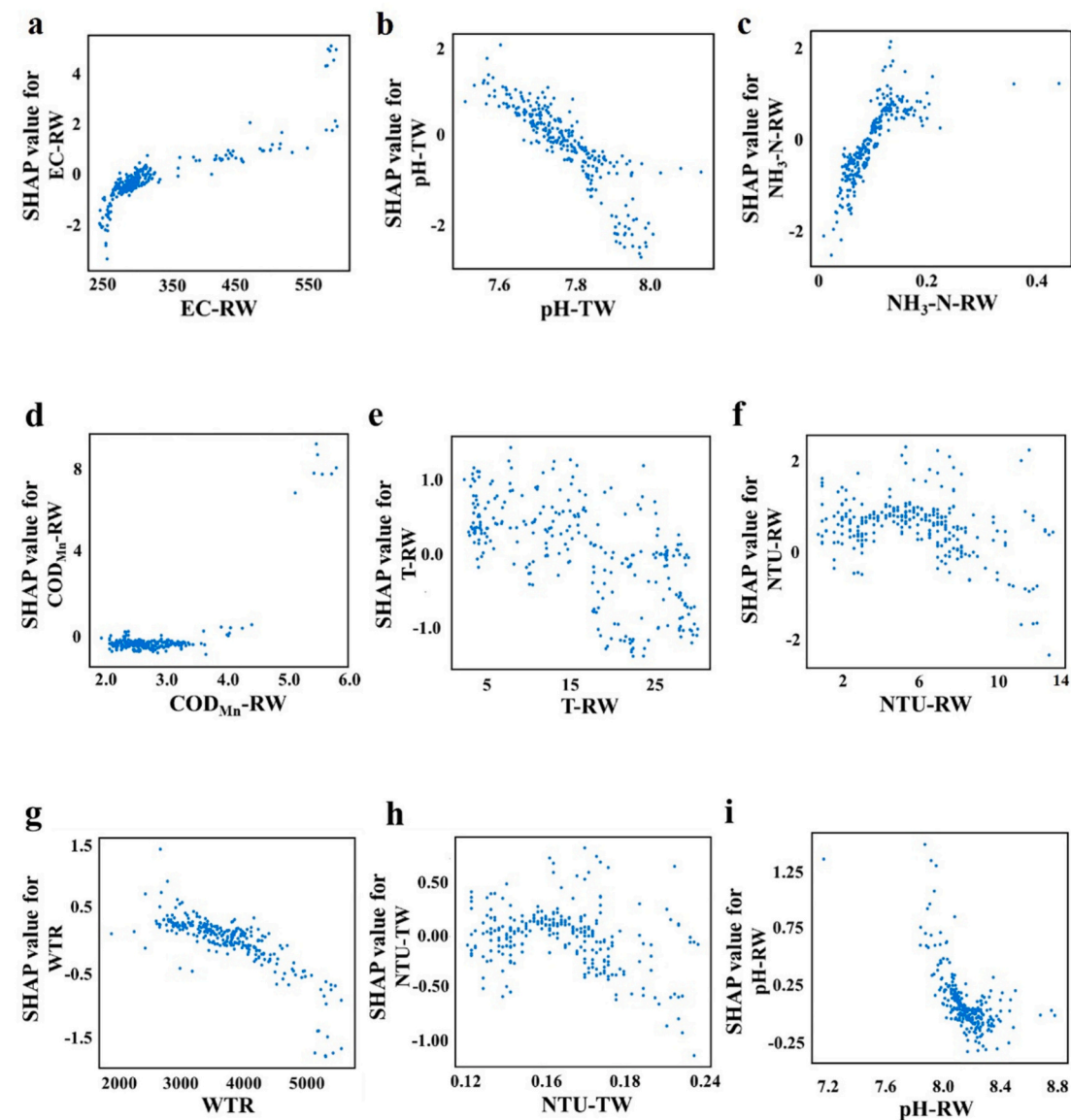


Fig. 7. Nonlinear relationships between individual water quality parameters and coagulant dosage for EC-RW (a), pH-TW (b), NH₃-N-RW (c), COD_{Mn}-RW (d), T-RW (e), NTU-RW (f), WTR (g), NTU-TW (h), and pH-RW (i). For each data point with a positive SHAP value, the dosage of coagulant increases with the SHAP value, exerting a positive impact. Conversely, for points with negative SHAP values, the opposite effect is observed. The units for these variables are consistent with those in Fig. 2.

hydrolysis reaction of PACl releases H⁺, resulting in a decrease in effluent pH [59]. However, over-dosing chemicals to meet the effluent pH standard will cause the colloids to re-stabilize, resulting in a decrease in flocculation efficiency. In addition, the increase in dissolved aluminum residue can affect the safety of the effluent [54]. In addition, no apparent linear relationship is observed between PACl dosage and NTU-TW after coagulation, contrary to the results reported by Chiavola [60]. NTU-TW is affected by both raw water quality and coagulant dosage. Specifically, elevated NTU-RW values require a manual increase in coagulant dosage to meet NTU-TW standards. Therefore, the NTU-TW values from an operating plant remain relatively low and stable with minimal variation, which has little impact on dosage prediction. In addition, the effectiveness of PACl in the coagulation process depends primarily on charge neutralization and adsorption bridging. An appropriate dosage of PACl hydrolyzes to produce positively charged ions that neutralize negatively charged colloids, thereby enhancing turbidity removal. However, excessive PACl triggers “colloidal protection,” where excess positively charged ions adsorb onto colloidal surfaces and

hydrophobic colloids become hydrophilic, reducing floc settling and ultimately decreasing turbidity removal efficiency [56,61,62].

Although AutoML (TPOT) and SHAP perform well in optimizing machine learning models and improving interpretability, they still have several limitations. AutoML may face the risk of overfitting during model selection and hyperparameter optimization, and in this study, despite the training set $R^2 = 1.00$, the high consistency of the repeated cross-validation and the independent test set ($R^2 = 0.96$) suggests that the model was not overfitted. The SHAP method reveals complex action laws that cannot be captured by traditional linear methods, identifies the threshold response behavior of key parameters, and breaks through the limitations of linear statistics.

Fig. 8 shows waterfall plots and decision plots for two specific local samples using the SHAP method. Fig. 8a and b show a sample with a coagulant dosage higher than the average value (14.5 mg/L). This sample was collected in February, with the raw water coming from the Luanhe River, and T-RW recorded at 3.8 °C. Fig. 8a shows that the predicted coagulant dosage for this sample (34.9 mg/L) is higher than

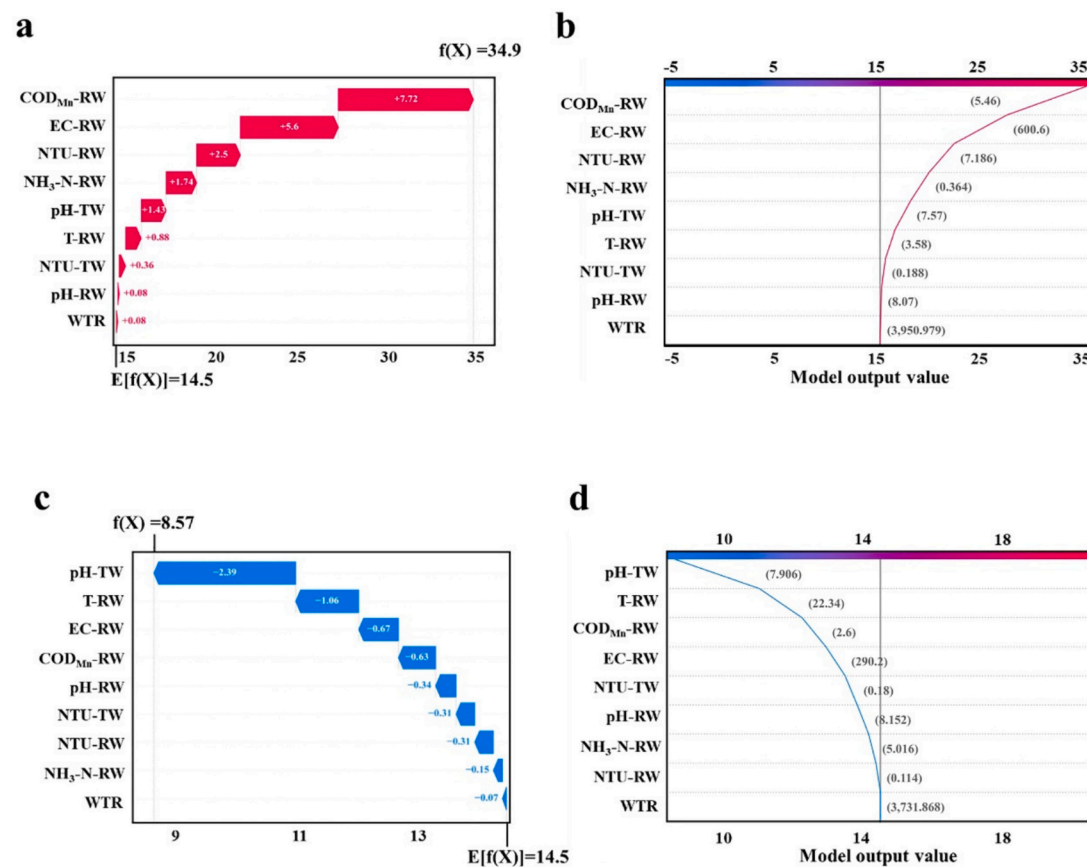


Fig. 8. Decision plots and waterfall plots for two local samples, featuring three key characteristics: 1) the output value, representing the prediction for an individual sample; 2) the base value, corresponding to the average expected value in the testing set; and 3) colour coding, where red variables push the prediction upwards and blue variables push it downwards) are presented. The Fig includes decision plot (a) and waterfall plot (b) for a sample with dosing amount above the average, and decision plot (c) and waterfall plot (d) for a sample with dosing amount below the average. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the average expected value in the test set (14.5 mg/L), mainly due to COD_{Mn}-RW, followed by EC-RW, NTU-RW, and NH₃-N-RW, while the contribution of pH-RW and WTR are negligible. Fig. 8b shows the contribution of each feature in exceeding the average prediction (14.5 mg/L), ultimately resulting in a final output of 34.9 mg/L for this sample. First, COD_{Mn}-RW reaches 5.46 mg/L, exceeding the threshold of 4 mg/L, the SHAP value increases significantly. In addition, with EC-RW at 600.6 μ S/cm, NH₃-N-RW at 0.36 mg/L and NTU-RW at 7.19 NTU, their SHAP values all remain above 0. Based on professional judgment of the overall impact of raw water quality, the dosage of PACl was subsequently increased from the average level.

Fig. 8c and d show a sample taken in May with the Yangtze River as the source water, showing a coagulant dosage below the average. Fig. 8c shows that the predicted coagulant dosage for this sample (8.57 mg/L) is significantly lower than the average expected value in the test set (14.5 mg/L). Among the factors, pH-TW (with a value of 7.906) has the most significant negative effect on the dosage prediction, indicating that a coagulant dosage at this level will significantly lower the pH of the treated water. This is followed by T-RW, EC-RW, and COD_{Mn}-RW, while the contribution of NH₃-N-RW and WTR is negligible. Fig. 8d further illustrates how each feature contributes to shifting the dosage towards a value below the average prediction (14.5 mg/L), ultimately resulting in a final output of 8.57 mg/L for this sample. For the raw water characterized by T at 22.34 °C, EC at 290.2 μ S/cm, COD_{Mn}-RW at 2.6 mg/L, and pH at 8.15, the corresponding SHAP values for these parameters are uniformly negative. Based on practical experience, it was decided to reduce the dosage of PACl to a level below the average.

In summary, the model developed in this study optimizes coagulant

dosing, not only enabling precise chemical addition in the coagulation process of water treatment plants—thereby significantly minimizing coagulant overuse—but also employing interpretable methods to reveal key factors and their impact relationships influencing coagulant consumption. This empowers plant operators to make more scientific and rational decisions, avoiding overdosing while ensuring effluent quality. However, in practical water treatment applications, further efforts are required to mitigate the environmental impacts of coagulant use. This can be achieved by integrating the model with processes such as pre-chlorination and activated carbon adsorption to reduce coagulant demand. Additionally, developing green and sustainable water treatment technologies, such as bio-based flocculants (e.g., chitosan, microbial flocculants), photocatalytic technologies[63,64], could reduce reliance on non-renewable resources and minimize environmental risks posed by chemical residues and byproducts (e.g., aluminum hydroxide sludge).

4. Conclusions

In this study, we developed a predictive model for coagulant dosing in a water treatment plant using AutoML and interpreted the model results to address the transparency issue associated with such predictive models. The conclusions are as follows:

- (1) The RF model developed using AutoML demonstrated superior performance with a test R^2 value of 0.96, RMSE of 0.89, and MAE of 0.47. This model offers a faster development cycle and higher prediction accuracy compared to manually selected models, and

serves as a valuable tool for predicting coagulant dosing in water treatment plants.

- (2) Model interpretation based on SHAP values revealed the influence of each water quality indicator on the prediction results. Specifically, when NTU-RW is low and relatively stable, raw water conductivity emerges as the most significant factor influencing coagulant dosing, followed by $\text{NH}_3\text{-N}$ -RW, COD_{Mn} -RW, T-RW, and WTR.
- (3) The optimized model predicts daily PACl savings of 222 kg (11 %) for water from the Yangtze River and 225 kg (8 %) for water from the Luanhe River. This highlights the key role of the coagulant dosing model in improving water treatment efficiency and optimizing resource utilization.

CRedit authorship contribution statement

Liyan Feng: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ying Zhang:** Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation. **Xiaoting Wei:** Writing – original draft, Visualization, Supervision, Software. **Mengyuan Wang:** Visualization, Validation, Software, Resources, Project administration. **Zhiguang Niu:** Resources, Project administration, Methodology, Investigation, Data curation. **Chenchen Wang:** Writing – review & editing, Visualization, Supervision, Software, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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